



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G01

SYSTEMS INTEGRATION AND ARCHITECTURE 1

FINALIZED

SOFTWARE REQUIREMENTS

SPECIFICATION (SRS)

Project Title: WashTrack: Laundry Pickup and Order Tracking System
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Date of Submission: June 30, 2026
Version: 2.0

Revision History

Version	Date	Description of Changes
1.0	June 22, 2026	Initial SRS submitted — project background, problem statement, preliminary functional and non-functional requirements, and partial scope defined.
2.0	June 30, 2026	Finalized SRS: restructured introduction; added Stakeholders section; assigned FR/NFR IDs; added Business Rules; added Requirements Traceability Table; added all UML diagrams (Use Case, ERD, Activity, Sequence). Clarified and expanded all requirements to be specific and testable.

1. Introduction

1.1 Project Title

WashTrack: Laundry Pickup and Order Tracking System

1.2 System Overview

WashTrack is a multi-platform digital system (mobile and web) that enables laundry businesses to digitize their entire order lifecycle — from customer pickup scheduling to staff-managed order processing and real-time status notifications. Payment is handled in-person at the physical laundry shop; the system displays the estimated amount due to the customer and allows staff to mark an order as paid once payment is collected. The system replaces manual paper-based record-keeping with a centralized, real-time platform.

1.3 Purpose of the System

The purpose of WashTrack is to eliminate inefficiencies in traditional laundry shop operations by providing:

- A customer-facing interface for submitting orders and tracking progress.
- A staff-facing interface for managing, updating, and processing orders.
- An automated notification system that keeps customers informed at every stage.
- A centralized database ensuring accurate and auditable records.

1.4 Target Users

- Customers — Individuals who use the laundry service.
- Staff — Laundry shop employees who manage and process orders.

1.5 Problem Statement

Laundry businesses commonly rely on manual recording methods such as notebooks and verbal communication, which result in lost or inaccurate records, delayed status updates, and customer dissatisfaction. Customers are required to contact the shop directly to inquire about order progress. Staff spend excessive time on manual record management instead of service delivery. WashTrack addresses these problems by providing a centralized, real-time digital platform.

1.6 Scope of the System

WashTrack will provide the following capabilities:

- User registration, login, and profile management
- Laundry pickup scheduling with date, time, and address selection
- Laundry order creation with service type, weight, and special instructions
- Real-time order tracking with status progression
- Staff order management and status update interface
- In-person payment recording by staff (cash only; no online payment gateway)
- Estimated price display for customers before and after order processing
- Automated push/in-app notifications on order status changes
- Order history viewing for customers

The following are explicitly out of scope:

- GPS vehicle tracking or automated route optimization
- Inventory management
- Multi-branch management
- Integration with third-party courier services
- AI-powered laundry care recommendations

1.7 Assumptions and Constraints

Assumptions

- All users have a smartphone or computer with a stable internet connection.
- Customers possess a valid email address for account registration.
- Staff regularly log in and update order statuses throughout the business day.
- The system is deployed on a reliable cloud-hosted server.

Constraints

- The system must be developed using Android (Kotlin) for mobile and a web framework for the web application.
- The database is MySQL and must be hosted on a server accessible 24/7.
- The system does not include real-time GPS tracking; pickup scheduling is time-slot based.

- Payment is collected in person at the physical laundry shop only. No online payment gateway is included. Customers may pay either upon delivery of their laundry or upon pickup of their completed order.

Stakeholders and Users

Customer

- Creates laundry orders
- Tracks order status
- Views order history
- Receives notifications

Staff

- Manages customer orders
- Updates order status
- Records payments

3. Functional Requirements

The following table lists the functional requirements of WashTrack. Each requirement is specific, testable, and traceable to at least one UML model.

Req. ID	Feature Name	Requirement Description	Priority
FR-001	User Registration	The system shall allow new users to create an account by providing full name, email, phone number, address, and password. The email address must be unique per account.	High
FR-002	User Login	The system shall authenticate users by verifying their email and password. Upon successful login, a session token shall be generated and the user redirected to their dashboard.	High
FR-003	Profile Management	The system shall allow authenticated customers to view and update their profile information, including name, phone number, and address.	Medium
FR-004	Schedule Pickup Request	The system shall allow customers to schedule a laundry pickup by selecting a date, time slot, and pickup address. The scheduled date must not be in the past.	High
FR-005	Create Laundry Order	The system shall allow customers to create a laundry order by specifying the service type (wash, dry-clean, fold), estimated weight in kilograms, and optional special instructions.	High
FR-006	Real-Time Order Tracking	The system shall display the current status of a customer's order using a defined set of statuses: Pending, Picked Up, Processing, Ready, Delivered. Status must update within 30 seconds of a staff change.	High
FR-007	Staff: Update Order Status	The system shall allow authenticated staff members to change the status of any order through the defined status progression. Each status change must be timestamped and logged.	High
FR-008	View Estimated Price	The system shall display the estimated total price on the customer's order detail screen once staff has recorded the weight and finalized the order. No payment is made through the system.	Medium
FR-009	Staff: Record Payment	The system shall allow staff to mark an order as paid after the customer pays in person at the shop. Staff shall record	High

Req. ID	Feature Name	Requirement Description	Priority
		whether payment was collected on laundry delivery or on customer pickup of finished laundry.	
FR-010	Notification System	The system shall automatically send an in-app and/or push notification to the customer whenever their order status changes. Notifications must be logged with a timestamp.	High
FR-011	View Order History	The system shall allow customers to view a paginated list of their past and current orders, sorted by creation date descending, with order ID, status, and total price displayed.	Medium
FR-012	Logout	The system shall allow authenticated users to log out, which shall invalidate their session token and redirect them to the login page.	Low

4. Non-Functional Requirements

Req. ID	Category	Requirement Description
NFR-001	Performance	The system shall load any page or screen within 3 seconds under normal network conditions (≥ 10 Mbps). API responses shall return within 2 seconds for 95% of requests.
NFR-002	Scalability	The system shall support a minimum of 100 concurrent users without degradation in response time.
NFR-003	Security – Authentication	All user passwords shall be hashed using bcrypt with a minimum cost factor of 10 before storage. Plaintext passwords shall never be stored.
NFR-004	Security – Authorization	The system shall enforce role-based access control. Customers shall not access staff-only routes or features. Staff shall not access customer account data beyond what is needed to manage orders.
NFR-005	Usability	The system's primary user workflows (login, create order, track order) shall be completable in no more than 4 screen interactions by a first-time user.
NFR-006	Reliability	The system shall maintain 99% uptime monthly. All data modifications shall be atomic and transactional to prevent partial updates.
NFR-007	Compatibility	The mobile application shall be compatible with Android 8.0 (API level 26) and above. The web application shall be compatible with Chrome, Firefox, Edge, and Safari latest versions.
NFR-008	Maintainability	The system codebase shall follow MVC architecture, with service and data access layers clearly separated. All modules shall be documented with inline comments.

5. Business Rules

The following business rules define the policies and constraints that govern system behavior.

Rule ID	Rule Name	Description
BR-001	Account Uniqueness	Each email address may be registered to only one account in the system. Duplicate email registrations shall be rejected.
BR-002	Pickup Scheduling	A pickup request must be scheduled at least 24 hours in advance. Same-day scheduling is not permitted.
BR-003	Order Status Progression	Order status must follow the defined sequence: Pending → Picked Up → Processing → Ready → Delivered. Status cannot be skipped or reversed.
BR-004	In-Person Payment Only	The system shall not process or collect any payment online. All payments are collected by staff in person at the physical shop. The system only records whether an order has been paid and when.
BR-005	Notification Trigger	The system shall automatically generate a notification whenever an order's status is changed by staff. Manual notifications by staff are also permitted.
BR-006	Role-Based Access	A user account is assigned exactly one role (Customer or Staff) at registration or account creation. Role changes require manual action by the shop owner.
BR-007	Data Retention	Order records and associated payment status records shall be retained in the system for a minimum of 1 year after the order's completion date.
BR-008	Weight Validation	An order's estimated weight must be greater than 0 kg. A maximum single-order weight of 50 kg shall be enforced.

6. Requirements Traceability Table

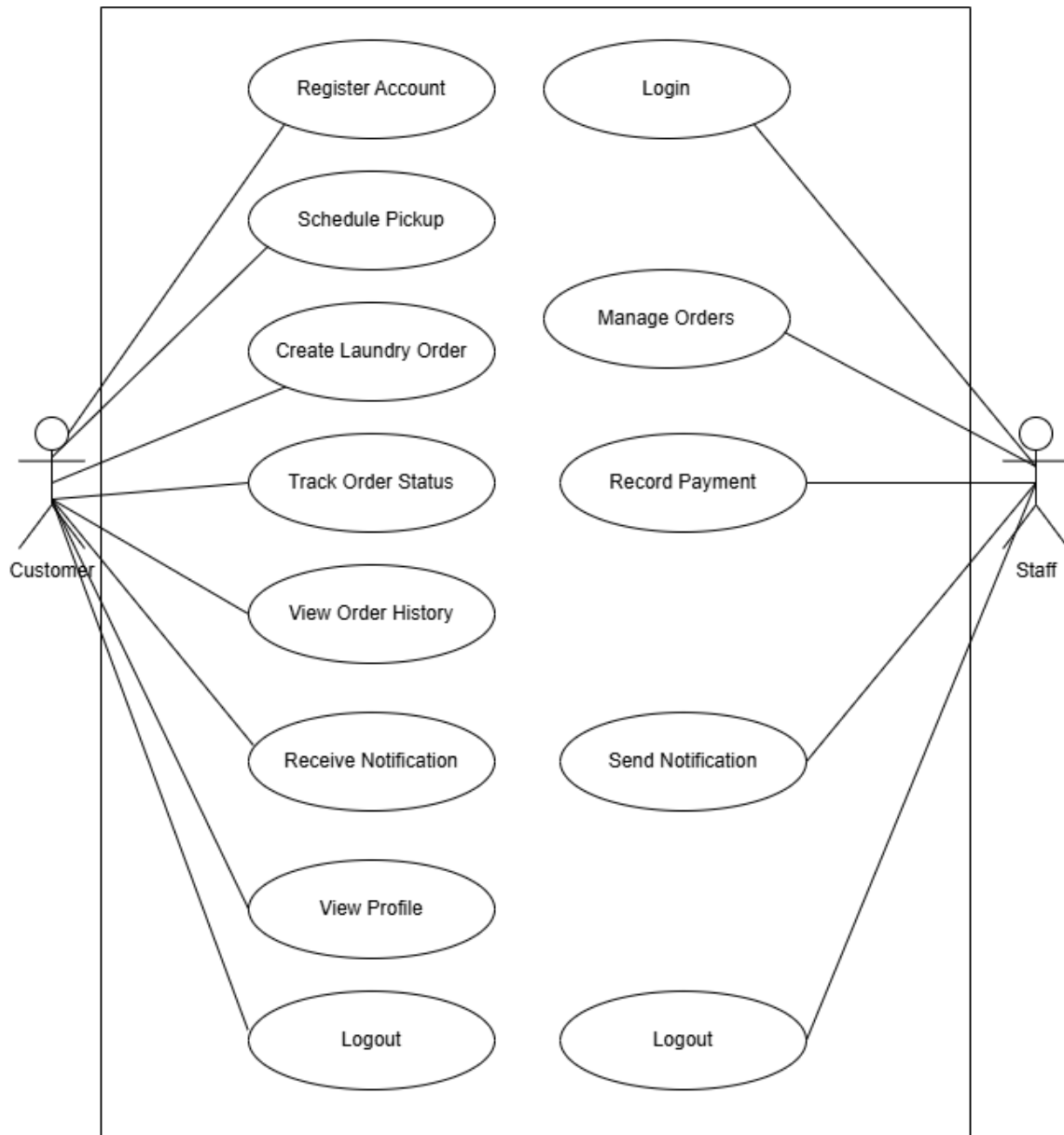
The following table maps each functional requirement to its corresponding use case name and the UML models that document it.

Req. ID	Feature Name	Use Case Name	UML Models
FR-001	User Registration	Register Account	Use Case Diagram, Sequence Diagram, ERD (USER)
FR-002	User Login	Login	Use Case Diagram, Sequence Diagram
FR-003	Profile Management	View Profile	Use Case Diagram, ERD (USER)
FR-004	Schedule Pickup	Schedule Pickup	Use Case Diagram, Activity Diagram, ERD (PICKUP_REQUEST)
FR-005	Create Laundry Order	Create Order	Use Case Diagram, Activity Diagram, ERD (ORDER)
FR-006	Order Tracking	Track Order Status	Use Case Diagram, Activity Diagram, ERD (ORDER)
FR-007	Update Order Status	Manage Orders	Use Case Diagram, ERD (ORDER), Activity Diagram
FR-008	View Estimated Price	View Order Details	Use Case Diagram, ERD (ORDER)
FR-009	Staff: Record Payment	Record Payment	Use Case Diagram, ERD (PAYMENT)
FR-010	Notification System	Receive Notifications	Use Case Diagram, ERD (NOTIFICATION), Sequence Diagram
FR-011	View Order History	View Order History	Use Case Diagram, ERD (ORDER)
FR-012	Logout	Logout	Sequence Diagram

Required UML Models

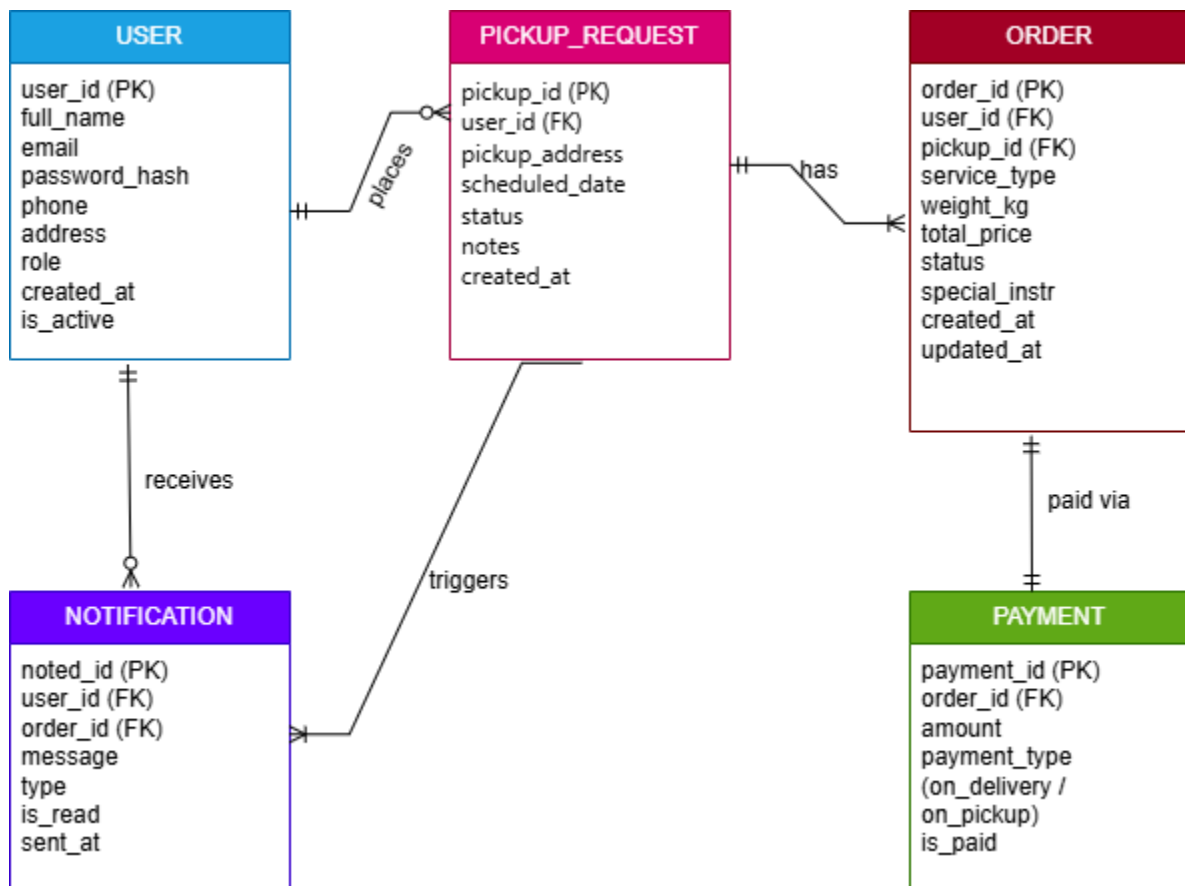
Use Case Model

The Use Case Diagram below shows all system actors (Customer and Staff) and their interactions with WashTrack. The system boundary encloses all use cases. The «include» relationship between Update Order Status and Send Notification indicates that a notification is always sent when an order status changes.



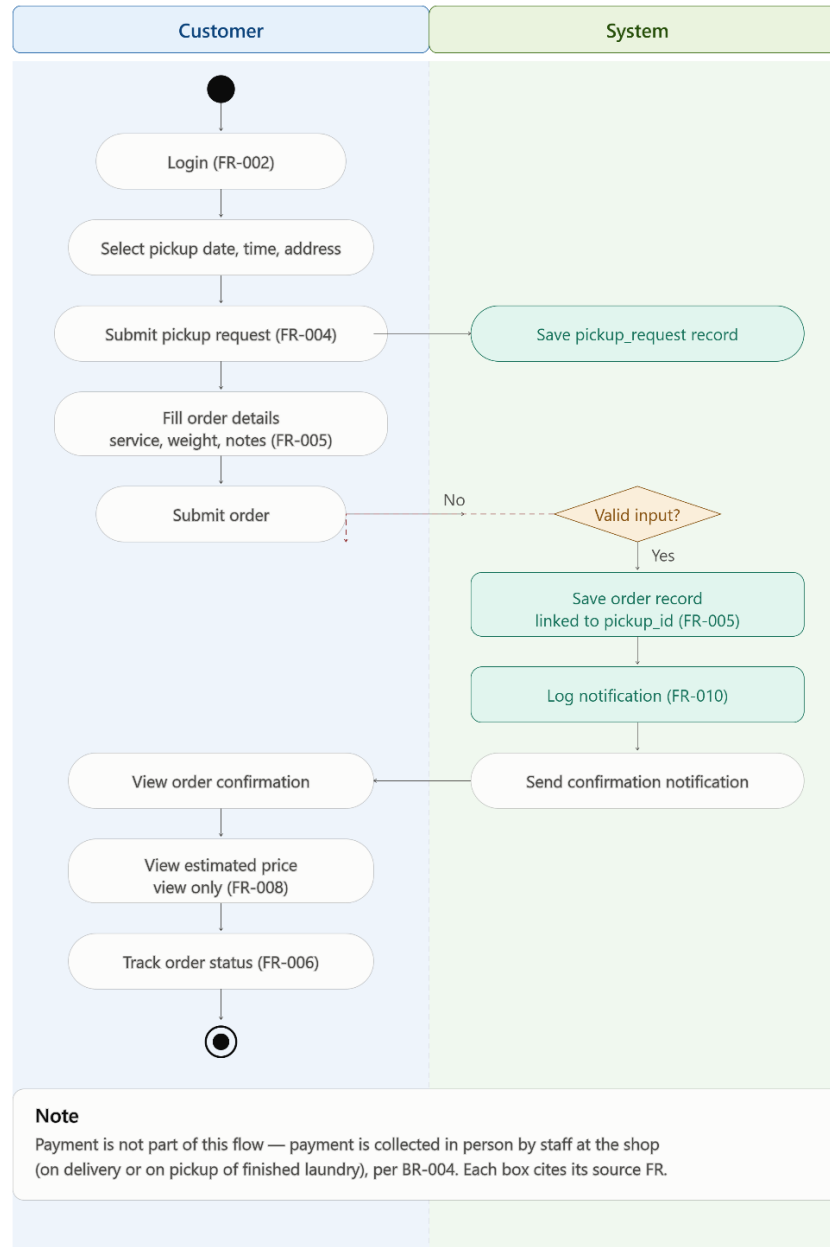
Entity Relationship Diagram (ERD)

The ERD below depicts the five core entities of WashTrack: USER, PICKUP_REQUEST, ORDER, PAYMENT, and NOTIFICATION. The PAYMENT entity records in-person payment status only — no financial transactions occur through the system. Primary keys (PK) are shown in red and foreign keys (FK) in blue. Cardinality notations (1 and N) are shown at relationship endpoints.



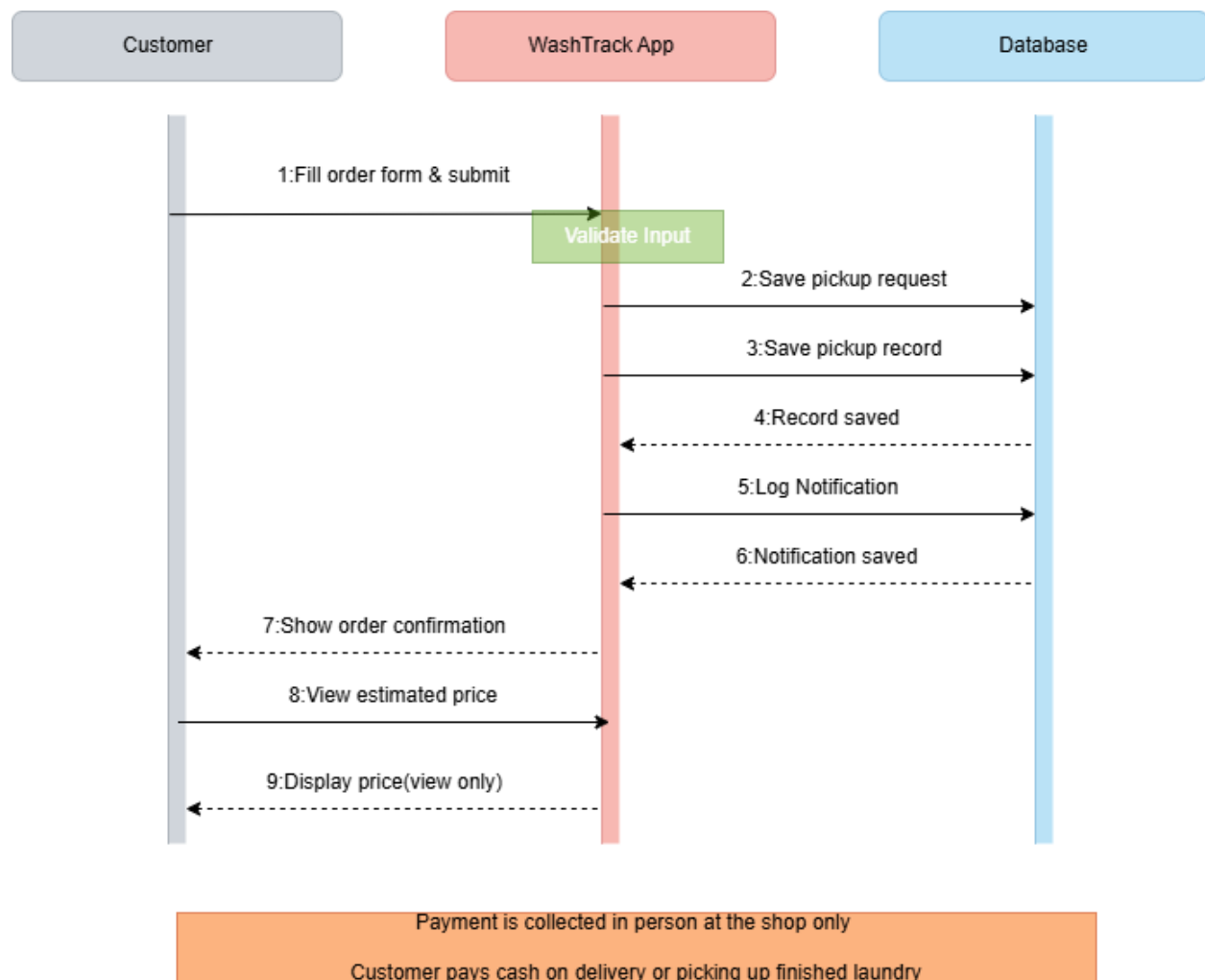
Activity Diagram – Order Placement

The Activity Diagram below illustrates the end-to-end workflow when a customer places a laundry order. The diagram uses swimlanes to separate Customer actions from System/Staff actions. A decision point validates the submitted form before a record is created. Synchronization bars indicate parallel completion points.



Sequence Diagram – Place Laundry Order

The Sequence Diagram below shows the interaction between the Customer, the WashTrack App, and the Database when placing a laundry order. After the customer submits the order form, the app saves the order and pickup records, logs a notification, and returns an order confirmation. The customer can then view the estimated price — for reference only. A note at the bottom of the diagram highlights that all payment is collected in person at the shop.



"I certify that this finalized SRS and UML models are my own individual work. I understand that copied, duplicated, or AI-generated submissions without proper understanding and revision may be subject to verification and possible deductions."